Code No: 111AB **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B.Tech I Year Examinations, June - 2022
MATHEMATICS - I
(Common to CE, EEE, ME, ECE, CSE, CHEM, EIE, IT, MCT, AE, AME, MIE, PTM, AGE)

Time: 3 Hours **Max. Marks: 75**

Answer any five questions
All questions carry equal marks
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1.a) Determine the rank of the matrix $A=\begin{bmatrix}1 & 3 & 4 & 3\\ 3 & 9 & 12 & 3\\ 1 & 3 & 4 & 1\end{bmatrix}$.

b) Investigate the values of λ and μ so that the equations $2x+3y+5z=9, 7x+3y-2z=8, 2x+3y+\lambda z=\mu$, have (i) no solution, (ii) a unique solution and (iii) an infinite number of solutions. [7+8]

2. Reduce the quadratic form $3x^2+5y^2+3z^2-2yz+2zx-2xy$ to the canonical form and also write the nature of the quadratic matrix. [15]

3.a) State Rolle's theorem and verify it for $f(x)=\frac{\sin x}{e^x}$ in $[0,\pi]$.

b) If $x=r\cos\theta, y=r\sin\theta$ then prove that $\frac{\partial(x,y)}{\partial(r,\theta)}\times\frac{\partial(r,\theta)}{\partial(x,y)}=1$. [7+8]

4.a) Prove that $\beta(m,n)=\frac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)}$.

b) Evaluate $\int_0^\infty\int_0^\infty e^{-(x^2+y^2)}dx\,dy$ by changing to polar coordinates. [8+7]

5.a) If a substance cools from 370 ^{0}k to 330 ^{0}k in 10 min., when the temperature of the surrounding air is 290 ^{0}k, find the temperature of the substance after 40 min.

b) Solve $\left(D^2+2\right)y=x^2+x^3+e^{-2x}+\cos 3x$, *where* $D=\frac{d}{dx}$. [7+8]

6.a) Solve $\left(D^3-D\right)y=2x+1+4\cos x+2e^x$, *where* $D=\frac{d}{dx}$.

b) Solve by method of variation of parameters $\frac{d^2y}{dx^2}+4y=\tan 2x$. [8+7]

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7.a) Find $L\left\{\frac{\sin 3t.\cos t}{t}\right\}$ and $L\left\{t^2\cos 2t\right\}$.

b) Find $L^{-1}\left\{\frac{s^2}{\left(s^2+4\right)\left(s^2+9\right)}\right\}$ Using Convolution theorem. [8+7]

8.a) (i) Find the Laplace Transform of $\left\{\left(\sqrt{t}+\frac{1}{\sqrt{t}}\right)^3\right\}$ and

(ii) Find $L\left\{t\,e^{-t}\sin 2t\right\}$.

b) Solve the differential equation by Laplace transforms method.
$y''+4y'+3y=e^{-t}$, $y(0)=y'(0)=1$. [8+7]

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